Lecture 1: AI/ML Foundations and Closed-Loop Control for Particle Accelerators

Physics Foundations, Neural Surrogates, Reinforcement Learning, and Bayesian Optimization

Chong Shik Park, Ph.D.

- Department of Accelerator Science, Korea University, Sejong 30019, Republic of Korea
- Center for Accelerator Research, Korea University, Sejong 30019, Republic of Korea

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Why AI in Modern Particle Accelerators?

Meeting stringent sub-micron stability tolerances while managing non-linear actuator dynamics

The Operational Challenge: High-Precision Orbit Stabilization

- **Extreme Beam Stability Tolerances:**

Modern synchrotron light sources, colliders, and free-electron lasers (FELs) require transverse orbit stabilization to within $< 10\%$ of beam size (sub-micron RMS tolerances).

- **High-Power Hadron Linacs (e.g., KOMAC, SNS, ESS):**

– Uncorrected trajectory offsets through RF accelerating cavities and focusing quadrupoles induce space-charge forces, emittance growth, and beam halo.
– Fractional beam losses exceeding 1 W/m ($\sim 10^{-5}$ of total beam power) cause severe structural radioactivation, prohibiting hands-on maintenance.

- **Limitations of Purely Classical Linear Control:**

– Analytical linear response matrices (ORM) ignore magnetic core saturation at high currents, inducing overshoot and limit cycles.
– Physical diagnostics carry 10 μm sensor noise floors; naive feedback amplifies measurement jitter into erratic actuator ripple.
– Multi-parameter coupling requires fast non-linear digital twins and adaptive closed-loop policies.

Integrated Morning Lectures & Afternoon Hackathon

Cohesive one-day curriculum connecting physics foundations directly to competitive simulation benchmarks

Program Schedule & Learning Objectives

1. Lecture 1: AI/ML Foundations & Closed-Loop Control (09:10 – 10:30)

- **Lecturer:** Chong Shik Park, Ph.D. (Korea University, Sejong)
- **Focus:** Transverse beam dynamics, Hill's equation, Orbit Response Matrix (ORM), non-linear cubic saturation, MLP surrogate modeling, SVD vs bounded inversion, RL with physics priors, and Bayesian optimization.

2. Lecture 2: Agentic AI for Accelerator Research (10:40 – 12:00)

- **Lecturer:** Thorsten Hellert, Ph.D. (Lawrence Berkeley National Laboratory / Osprey)
- **Focus:** LLM agents, dynamic Python API tool execution, automated scientific workflows, simulation orchestration, and human-in-the-loop operational safety guardrails.

3. Afternoon Mini-Hackathon: Beam Orbit Alignment Benchmark (13:30 – 17:00)

- 15-meter transport lattice, 12 BPMs, 6 steerers, 3 parallel challenge tracks (MLP, RL, BO).
- Real-time experiment tracking with Weights & Biases (W&B) and live global leaderboard.

Five Pillars of AI in Accelerators (Part 1: Surrogates & Optimization)

Foundational applications revolutionizing beam physics simulations and online tuning

Pillars 1 & 2: Fast Digital Twins & Autonomous Multi-Objective Tuning

Pillar 1: Neural Surrogate Digital Twins (Fast Forward Simulation)

- High-fidelity particle tracking codes (`ELEGANT`, `MAD-X`, `OPAL`) simulate detailed collective space-charge forces, but require minutes to hours per run.
- Deep Neural Networks (MLPs, PINNs) learn compact, high-accuracy representations capable of sub-millisecond execution times ($<$ 0.2 ms).
- **Operational Impact:** Enables real-time model-predictive control (MPC), online orbit feedback, fast virtual diagnostics, and parameter sensitivity scans.

Pillar 2: Autonomous Multi-Objective Optimization & Tuning

- Accelerator operations require simultaneously balancing conflicting objectives: maximizing transmission and photon brightness while minimizing emittance and beam halo.
- Bayesian Optimization (BO) with Gaussian Processes leverages epistemic uncertainty and exploration-exploitation acquisition functions.
- **Operational Impact:** Achieves optimal machine operating points directly on physical beamlines in dozens of shots rather than thousands of manual scan steps.

Five Pillars of AI in Accelerators (Part 2: Diagnostics, Protection & Agents)

Expanding AI applications into beam instrumentation, uptime preservation, and autonomous agents

Pillars 3, 4 & 5: Virtual Diagnostics, Protection & Agentic Control

Pillar 3: Virtual Diagnostics & Non-Destructive Phase Space Reconstruction

- Interceptive diagnostics (wire scanners, screens, deflecting cavities) destroy or disrupt the beam during measurement.
- Supervised ML reconstructs 6D phase space, bunch slice emittance, and current profiles shot-by-shot from non-invasive upstream RF probes and BPM readings.

Pillar 4: Anomaly Detection, Fault Prediction & Machine Protection

- Unscheduled downtime in multi-megawatt facilities incurs catastrophic financial and scientific operational costs.
- Unsupervised autoencoders detect precursor thermal drift; deep recurrent networks predict RF breakdown and superconducting cavity quenches milliseconds before failure.

Pillar 5: Agentic AI & Autonomous Operational Workflows

- Modern agent frameworks (e.g. *Osprey*) couple Large Language Model reasoning engines with standardized Python control APIs and simulation codes.
- Translates high-level natural language goals into validated, safe operational sequences under strict domain guardrails.

Transverse Beam Dynamics & Hill's Equation

The fundamental second-order differential equation governing transverse betatron motion

Derivation and Physical Interpretation of Hill's Equation

Governing Equation of Motion:

For an uncoupled linear magnetic transport line, particle displacement $x(s)$ along coordinate s satisfies:

$$\frac{d^2x(s)}{ds^2} + K(s)x(s) = 0$$

- **Quadrupole Focusing Function $K(s)$:**
  - $K(s) = \frac{e}{p}\frac{\partial B_y}{\partial x}$ defines the alternating gradient magnetic focusing strength.
  - $K > 0$ represents horizontally focusing quadrupoles (QF) that horizontally compress the beam envelope.
  - $K < 0$ represents horizontally defocusing quadrupoles (QD) that vertically focus the beam envelope.

- **Periodic Focusing Principle (Courant-Snyder Invariance):**

Even though $K(s)$ varies piecewise along the beam axis, alternating QF and QD pairs yield net focusing in both transverse planes simultaneously.

Betatron Oscillations, Beta Function & Betatron Tune

Analytical solution to Hill's equation and Courant-Snyder envelope parameters

Betatron Oscillation Solution & Envelope Dynamics

General Pseudo-Harmonic Betatron Solution:

The exact analytical solution to Hill's equation is expressed as a modulated harmonic wave:

$$x(s) = \sqrt{\epsilon \cdot \beta(s)} \cdot \cos\left(\psi(s) + \phi_0\right)$$

- **Physical Parameter Definitions:**
  - $\beta(s)$ [m]: Courant-Snyder beta function, describing the transverse beam envelope geometry along the lattice.
  - ϵ [m · rad]: Beam emittance, representing the conserved area occupied by the beam in transverse phase space (x, x').
  - $\psi(s) = \int_0^s \frac{d\bar{s}}{\beta(\bar{s})}$: The betatron phase advance accrued along the longitudinal coordinate.
  - $\nu = \frac{1}{2\pi} \oint \frac{ds}{\beta(s)}$: The betatron tune, representing the total number of transverse oscillations per machine turn.
- **Key Insight for Orbit Correction:**

Any localized angular deflection propagates downstream as a betatron oscillation modulated by $\sqrt{\beta(s)}$.

Derivation of the Orbit Response Matrix (ORM)

Mathematical formulation linking localized steering kicks to downstream BPM readings

The Analytical ORM Transfer Function

Closed-Orbit Perturbation from a Localized Steerer Kick:

When a steering dipole at position s_j applies an angular kick $\theta_j = \frac{eBL}{p}$, the orbit shift at BPM position s_i is:

$$R_{ij} = \frac{\Delta x(s_i)}{\theta_j} = \frac{\sqrt{\beta_i \beta_j}}{2\sin(\pi\nu)} \cos\left(|\psi_i - \psi_j| - \pi\nu\right)$$

- **Global Matrix Superposition Formulation:**

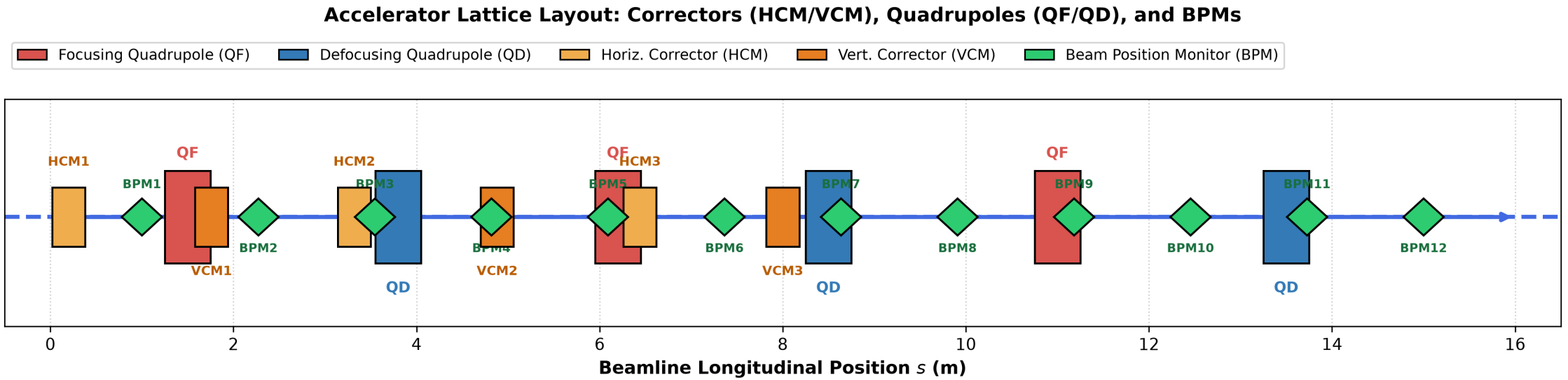
For N corrector magnets and M Beam Position Monitors, the linear response is encapsulated as:

$$\mathbf{Y}_{\text{linear}} = \mathbf{I} \cdot \mathbf{R}$$

– $\mathbf{I} \in \mathbb{R}^6$: Vector of corrector coil currents in Amperes, mapped to kicks via magnetic field integration.
– $\mathbf{R} \in \mathbb{R}^{6\times12}$: The linear Orbit Response Matrix (ORM).
– $\mathbf{Y} \in \mathbb{R}^{12}$: Transverse displacements measured at all 12 downstream BPM stations.
– Cross-plane coupling: Any single corrector deflection alters the entire beam trajectory downstream.

The 15-Meter Transport Lattice Architecture

Physical arrangement of quadrupole doublets, dipole correctors, and beam position monitors



Transport Line Specifications & Component Placement

- **12 Beam Position Monitors (BPMs):** Evenly distributed from $s = 1.0$ m to $s = 15.0$ m, measuring transverse displacement $\mathbf{Y}$ in millimeters.
- **6 Dipole Steering Magnets:** 3 Horizontal (HCM1-3) and 3 Vertical (VCM1-3) steerers positioned between $s = 0.2$ m and $s = 8.0$ m.
- **Alternating Quadrupole Doublets:** Pairs of QF and QD focusing magnets shaping the betatron phase advance and maintaining envelope stability.

Hardware Constraints & Actuator Bounds

Realistic power supply operational boundaries and sensor measurement jitter

Power Supply Limits and Diagnostic Noise Specifications

1. Cumulative Corrector Current Bounds (Thermal & Magnetic Saturation Limit):

- Total current state: $I_k \in [-5.0\text{ A}, +5.0\text{ A}]$ for all 6 correctors.
- Magnets entering currents $|I| > 3.0$ A experience pronounced iron core saturation.
- Hardware protection interlocks trip power supplies if current demands exceed ± 5.0 A.

2. Per-Step Delta Current Bounds (Inductive Slew Rate Limit):

- Single-step control action: $\Delta I_k \in [-0.8\text{ A}, +0.8\text{ A}]$.
- Prevents high inductive back-EMF spikes ($L \cdot \frac{di}{dt}$) that stress magnet coils and trigger power supply voltage limits.

3. Gaussian Diagnostic Measurement Noise (Sensor Resolution Floor):

- Sensor noise: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, with $\sigma = 0.010$ mm (10 μm RMS resolution).
- **Crucial for control design:** Naive controllers misinterpret sensor jitter as physical orbit errors, commanding false kicks.

Magnetic Core Saturation & Initial Distortion

Departing from idealized linear optics: modeling cubic saturation and quadrupole errors

Mathematical Formulation of Non-Linear Beamline Response

Complete Beam Position Response Formulation:

The actual BPM observation vector $\mathbf{Y}$ in the simulation environment is governed by:

$$\mathbf{Y} = \mathbf{I} \cdot \mathbf{R} - 0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R}) + \mathbf{Y}_0 + \boldsymbol{\epsilon}$$

- **Physical Breakdown of Response Terms:**

– $\mathbf{I} \cdot \mathbf{R}$: Ideal linear betatron wave response predicted by transfer matrix.
– $-0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R})$: Non-linear iron core soft saturation. At high excitation currents, the iron yoke reaches magnetic saturation, decreasing steerer kick efficiency by 6% (at 2 A) up to 24% (at 4 A).
– $\mathbf{Y}_0$: Initial orbit distortion caused by random quadrupole transverse misalignments (producing 0.40 mm to 1.10 mm RMS error across seeds).
– $\boldsymbol{\epsilon}$: Diagnostic Gaussian measurement jitter ($\sigma = 10$ μm).

- **Convergence Target:** **Drive orbit error below RMS < 0.030 mm (30 μm) within 20 control steps.**

Machine Learning Foundations: Artificial Neurons & Layers

Mathematical anatomy of feedforward perceptrons and non-linear activation functions

Neuron Transformation and Multi-Layer Composition

1. The Artificial Neuron Transformation:

Takes input vector $\mathbf{x} \in \mathbb{R}^d$, computes affine combination, and applies non-linear activation σ:

$$z = \sum_j w_j x_j + b = \mathbf{w}^T \mathbf{x} + b, \qquad a = \sigma(z)$$

2. Multi-Layer Perceptron (MLP) Deep Architecture:

Stacks L successive layers of affine transformations and non-linear activations:

$$\mathbf{h}^{(l)} = \sigma\left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right), \quad \textbf{with } \mathbf{h}^{(0)} = \mathbf{x}, \quad \hat{\mathbf{y}} = \mathbf{h}^{(L)}$$

3. Activation Function Selection: Rectified Linear Unit (ReLU):

- $\sigma(z) = \max(0, z)$: Yields constant derivative $= 1$ for $z > 0$, completely resolving the vanishing gradient problem inherent to sigmoid/tanh activations in deep networks.
- Computationally ultra-efficient, enabling fast backpropagation.

Loss Functions & The Adam Optimizer

Formulating surrogate objective functions and adaptive gradient descent mechanics

Objective Formulation and Adaptive Momentum Updates

Supervised Loss Formulation (Mean Squared Error + L2 Regularization):

To train a neural surrogate on offline simulation data $(\mathbf{I}_i, \mathbf{y}_{\text{true},i})$, we minimize the regularized loss:

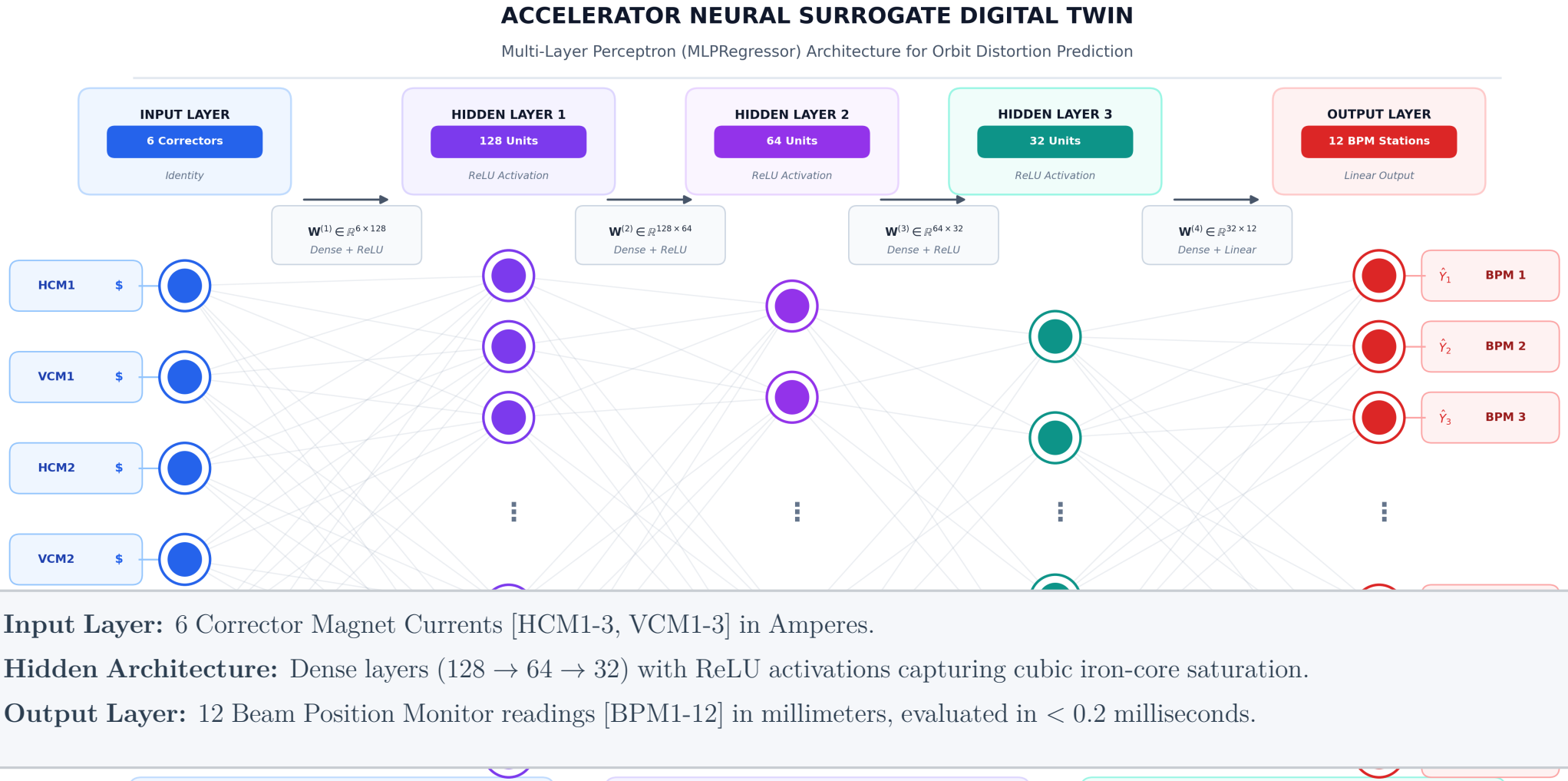
$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \Sigma_{i=1}^{N} \|\mathbf{y}_{\mathbf{true},i} - f_{\boldsymbol{\theta}}(\mathbf{I}_i)\|^2 + \lambda \|\boldsymbol{\theta}\|^2$$

The Adam Optimization Algorithm (Adaptive Moment Estimation):

- **First Moment $\mathbf{m}_t$:** Exponentially moving average of gradients (momentum):
  $\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1)\mathbf{g}_t, \quad (\beta_1 = 0.9)$
- **Second Moment $\mathbf{v}_t$:** Exponentially moving average of squared gradients (adaptive scale):
  $\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2)\mathbf{g}_t^2, \quad (\beta_2 = 0.999)$
- **Parameter Update with Bias Correction:**
  $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \hat{\mathbf{m}}_t$
- **Advantage for Accelerators:** Automatically adapts individual parameter steps across varying magnet coupling scales without tedious manual learning rate schedules.

Neural Surrogate Digital Twin Architecture

Deep Multi-Layer Perceptron (128-64-32) modeling non-linear beam response



- **Input Layer:** 6 Corrector Magnet Currents [HCM1-3, VCM1-3] in Amperes.
- **Hidden Architecture:** Dense layers ($128 \rightarrow 64 \rightarrow 32$) with ReLU activations capturing cubic iron-core saturation.
- **Output Layer:** 12 Beam Position Monitor readings [BPM1-12] in millimeters, evaluated in < 0.2 milliseconds.

1. Surrogate Architecture

Model: scikit-learn MLPRegressor
Layers: (6 -> 128 -> 64 -> 32 -> 12)
Activations: Hidden ReLU, Output Linear
Optimizer: Adam (learning rate = 1e-3)

2. Physical Domain Mapping

Inputs: 6 Corrector Currents [-3.0 A, +3.0 A]
Outputs: 12 BPM Trajectory Offsets (mm)
Core Non-Linearity: Cubic Iron Saturation
Simulation Noise: N(0, 0.010 mm) sensor jitter

3. Benchmark Performance

Linear Ridge Holdout RMSE: 0.0558 mm
Neural MLP Holdout RMSE: 0.0118 mm
Bayesian Sensor Noise Floor: 0.0100 mm
Forward Inference Latency: ~0.15 ms

Surrogate Dataset Generation & Holdout Verification

Ensuring digital twin fidelity across linear and saturated magnetic operating regimes

Offline Dataset Generation and Model Verification

1. Synthetic Tracking Dataset Generation:

- 2,000 synthetic simulation runs generated by uniformly sampling corrector currents across $\mathbf{I} \in [-3.0, +3.0]^6$ A.
- Full non-linear response computed including the cubic soft-saturation term $-0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R})$.
- Dataset partitioned into 80% Training (1,600 samples) and 20% Holdout Testing (400 samples).

2. MLPRegressor Training Configuration:

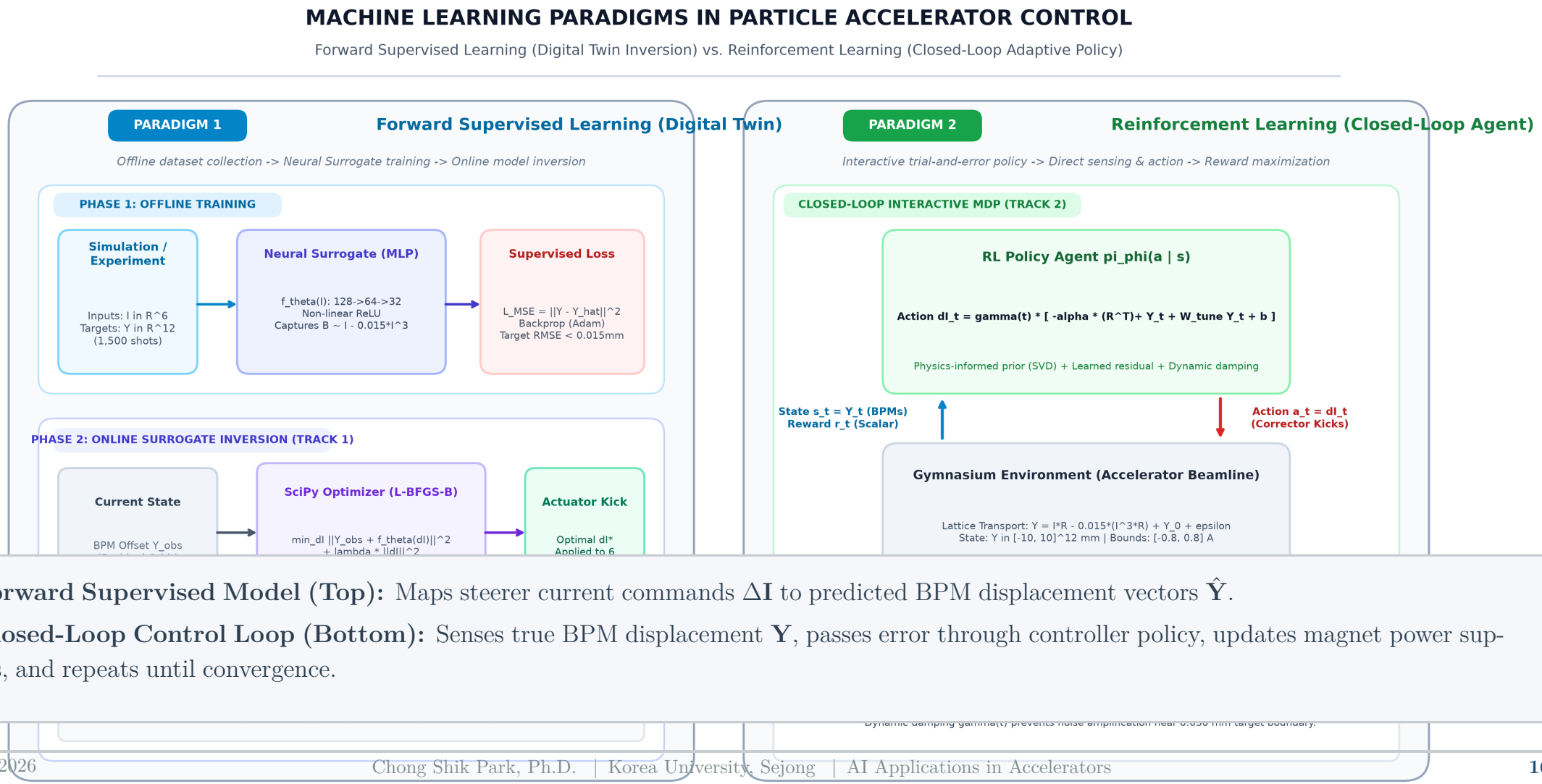
- Architecture: `hidden_layer_sizes=(128, 64, 32)`, `activation='relu'`, `solver='adam'`.
- Regularization: $\alpha = 10^{-4}$, `max_iter=500`, `early_stopping=True`.
- Loss convergence reached in under 180 epochs ($\sim$ 3.5 seconds CPU training time).

3. Holdout Verification Metrics:

- **Holdout Test MSE:** $< 2.5 \times 10^{-4}$ mm^2 across all 12 BPM stations.
- **Sub-Millisecond Execution:** Forward inference takes < 0.2 ms per call, 10,000$\times$ faster than numerical tracking.

Forward Digital Twin vs Closed-Loop Feedback Flow

Transitioning from predictive surrogate modeling to active closed-loop trajectory correction



- **Forward Supervised Model (Top):** Maps steerer current commands $\Delta\mathbf{I}$ to predicted BPM displacement vectors $\hat{\mathbf{Y}}$.
- **Closed-Loop Control Loop (Bottom):** Senses true BPM displacement $\mathbf{Y}$, passes error through controller policy, updates magnet power supplies, and repeats until convergence.

Baseline 0: Classical Linear SVD Controller

Inverting the linear Orbit Response Matrix via Moore-Penrose pseudo-inverse

Mathematical Formulation & Operational Vulnerabilities

Singular Value Decomposition (SVD) Pseudo-Inverse Formulation:

Computes corrective kick vector $\Delta\mathbf{I}$ directly by inverting the transpose of the linear ORM:

$$\Delta\mathbf{I} = -\alpha \cdot (\mathbf{R}^T)^\dagger \cdot \mathbf{Y}_{\mathbf{obs}}, \qquad \textbf{with } (\mathbf{R}^T)^\dagger = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^T$$

- **Operational Profile & Limitations:**

– Static feedback gain: Typically $\alpha \approx 0.45$, with kicks hard clipped to $[-0.8, +0.8]$ A.
– Execution latency: < 0.1 ms (instantaneous analytical matrix-vector product).
– **Vulnerability in Non-Linear Regimes:** Completely oblivious to cubic iron core saturation. When magnets operate at high currents, the true response is much weaker than linear $\mathbf{R}$ predicts, leading to over-correction, slow convergence (5.6 mean steps), and high final current demands.

- **Tested Benchmark Performance: 100.0% success rate, 5.6 mean steps, 1,013.1 total score (Rank #3).**

Baseline 1: Supervised Neural Surrogate Inversion

Model-based bounded optimization via SciPy L-BFGS-B on the trained neural digital twin

Optimization Formulation on Forward Surrogate

Per-Step Bounded Numerical Optimization Problem:

Instead of inverting an analytical matrix, Baseline 1 leverages the non-linear MLP twin $f_{\boldsymbol{\theta}}$:

$$\min_{\Delta \mathbf{I}} \mathcal{L}(\Delta \mathbf{I}) = \|\mathbf{Y}_{\mathbf{obs}} + f_{\boldsymbol{\theta}}(\Delta \mathbf{I})\|^2 + \lambda \|\Delta \mathbf{I}\|^2 \quad \textbf{subject to } \Delta \mathbf{I} \in [-5.0, +5.0]^6 \ \mathbf{A}$$

- **Why Supervised Inversion Outperforms Classical SVD:**
  - Because $f_{\boldsymbol{\theta}}$ captures the cubic saturation falloff, the optimizer automatically calculates higher current demands for saturated correctors.
  - Solver: SciPy `L-BFGS-B` converges in 8–15 iterations per control step ($\sim 15 - 25$ ms latency).
  - **Result:** Slashes convergence time from 5.6 steps (SVD) down to only 2.3 steps average!
- **Tested Benchmark Performance: 100.0% success rate, 2.3 mean steps, 1,016.8 total score (Rank #2).**

Closed-Loop RL with Physics-Informed Priors

Continuous policy networks combining analytical optics priors with learned residual offsets

The Physics-Informed Residual Architecture

Hybrid Residual Policy Formulation:

Rather than learning control from scratch with unguided exploratory random kicks, our agent uses:

$$\Delta \mathbf{I}(t) = \gamma(t) \cdot \left[-\alpha (\mathbf{R}^T)^{\dagger} \mathbf{Y} + \mathbf{W}_{\mathbf{learned}} \mathbf{Y} + \mathbf{b} \right]$$

- **Architectural Synergy Breakdown:**

1. **Analytical Physics Prior ($-\alpha(\mathbf{R}^T)^{\dagger}\mathbf{Y}$):**
   – Anchors the policy in first-principles beam optics.
   – Guarantees immediate coarse stabilization from step 1 without needing millions of trial steps.
2. **Learned Residual Offset ($\mathbf{W}_{\mathbf{learned}}\mathbf{Y} + \mathbf{b}$):**
   – Neural network learns to predict non-linear saturation compensation.
   – Balances asymmetric horizontal (HCM) and vertical (VCM) steerer authorities dynamically.
3. **Execution Latency:** < 0.4 ms (pure feedforward matrix multiply, $50\times$ faster than iterative `L-BFGS-B`).

Dynamic Harmonic Step Damping

Preventing sensor-noise amplification and eliminating limit-cycle oscillations

Resolving the Sensor Noise Feedback Limit Cycle

The Noise Floor Bottleneck:

- As the orbit converges toward the sub-30 μm target, true orbit errors become comparable to BPM sensor noise ($\sigma = 10$ μm).
- Undamped linear feedback interprets stochastic sensor jitter as real trajectory error, commanding vigorous kicks that induce limit cycles around the boundary.

Harmonic Damping Schedule:

We modulate the policy kick authority with a dynamic temporal decay schedule:

$$\gamma(t) = \frac{1.0}{1.0+\kappa \cdot t}, \qquad \textbf{with } \kappa \approx 0.03$$

- **Step 1** ($t = 0, \gamma = 1.0$)**:** Full kick authority rapidly suppresses large millimeter-scale initial distortions.
- **Step 5+** ($t \geq 4, \gamma < 0.88$)**:** Attenuated feedback prevents sensor jitter amplification, locking the orbit cleanly below 0.030 mm.

Closed-Loop Bayesian Optimization with Gaussian Processes

Sample-efficient probabilistic modeling with epistemic uncertainty estimation

Probabilistic Gaussian Process Modeling

Gaussian Process Surrogate Formulation:

Instead of deterministic point predictions, a GP models a distribution over candidate orbit response functions:

$$\hat{\mathbf{Y}}(\Delta\mathbf{I}) \sim \mathcal{GP}\left(\mathbf{m}(\Delta\mathbf{I}), \mathbf{K}(\Delta\mathbf{I}, \Delta\mathbf{I}')\right)$$

- **Composite Physics-Informed Kernel Architecture:**

$$k(\mathbf{x}, \mathbf{x}') = k_{\textbf{Matérn-5/2}}(\mathbf{x}, \mathbf{x}') + \sigma^2_{\textbf{noise}} \cdot \delta(\mathbf{x}, \mathbf{x}')$$

– **Matérn-5/2 Kernel:** Captures smooth, non-linear magnetic saturation falloff without imposing the unrealistic infinite differentiability of RBF kernels.
– **White Noise Kernel:** Explicitly accounts for the 10 μm BPM diagnostic jitter floor, avoiding kernel overfitting to noise.

- **Epistemic Uncertainty Estimation:**

For any candidate kick $\Delta\mathbf{I}$, the GP provides both posterior mean $\boldsymbol{\mu}_{\text{GP}}(\Delta\mathbf{I})$ and standard deviation $\boldsymbol{\sigma}_{\text{GP}}(\Delta\mathbf{I})$.

Lower Confidence Bound (LCB) Acquisition & Optimization

Balancing exploration of non-linear saturation against exploitation of predicted orbit corrections

The LCB Acquisition Strategy

Lower Confidence Bound (LCB) Mathematical Formulation:

At each step, the controller selects kicks by minimizing the LCB acquisition criterion:

$$\alpha_{\mathrm{LCB}}(\Delta\mathbf{I}) = \mathbf{RMS}\left(\mathbf{Y}_{\mathrm{obs}} + \boldsymbol{\mu}_{\mathrm{GP}}(\Delta\mathbf{I})\right) - \kappa \cdot \overline{\boldsymbol{\sigma}}_{\mathrm{GP}}(\Delta\mathbf{I}) + \lambda\|\Delta\mathbf{I}\|^2$$

- **Parameter Interpretations:**

– $\boldsymbol{\mu}_{\mathrm{GP}}(\Delta\mathbf{I})$: Posterior mean predicted orbit deviation across all 12 BPM stations.
– $\overline{\boldsymbol{\sigma}}_{\mathrm{GP}}(\Delta\mathbf{I})$: Posterior epistemic uncertainty (average standard deviation across BPMs).
– $\kappa \geq 0$: Exploration coefficient. When $\kappa > 0$, the agent explores uncharted high-current regimes; when $\kappa = 0$, it exploits the calibrated posterior mean.
– $\lambda\|\Delta\mathbf{I}\|^2$: Regularization penalty that minimizes actuator heating and avoids saturation.

- **Exceptional Convergence Speed:** **When calibrated, Track 3 reaches the convergence threshold in an average of only 1.5 steps!**

Gymnasium Environment Specification

Standard Farama Gymnasium API contract governing state, actions, and rewards

The Farama Gymnasium Environment Interface

Standard Gymnasium Method Signatures:

```
obs, info = env.reset(seed=seed)
obs, reward, terminated, truncated, info = env.step(action)
```

- **Observation Space:** `Box(-10.0, +10.0, shape=(12,), dtype=float32)` (12 transverse BPM readings in mm).
- **Action Space:** `Box(-0.8, +0.8, shape=(6,), dtype=float32)` (6 per-step delta kicks $\Delta\mathbf{I}$ in Amperes).
- **Step Reward Function:**

$$r_t = -\mathbf{RMS}_{t+1} - 0.01 \cdot \|\Delta\mathbf{I}\|_2 + (50.0 \textbf{ if } \mathbf{RMS}_{t+1} < 0.030 \textbf{ mm else } 0.0)$$

- **Termination Criteria:** `terminated = True` when RMS < 0.030 mm (success); `truncated = True` when step ≥ 20 (budget exhausted).

Standardized 30-Seed Scoring Formula

Multi-objective evaluation metric balancing convergence reliability, speed, precision, and power

The Official Hackathon Scoring Formula

Standardized Evaluation Suite: `seeds = range(1, 31)` (30 evaluation seeds)

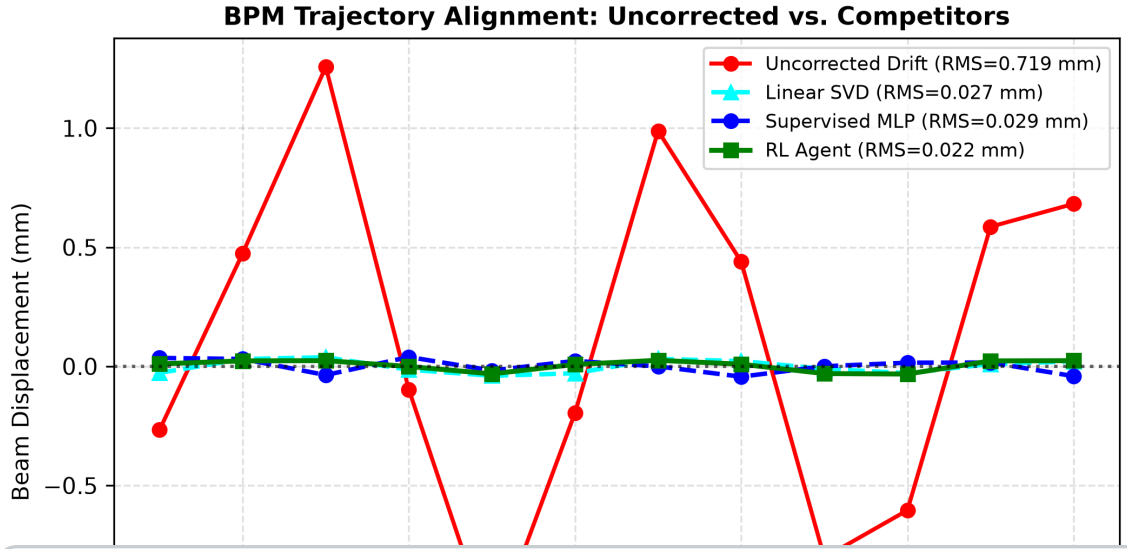
$$\textbf{Score} = (\textbf{Success Rate} \times 10.0) - (2.0 \times \overline{\textbf{Steps}}) + \left(\frac{0.03}{\max(\overline{\textbf{RMS}}, 0.005)} \times 20.0\right) - (2.0 \times \overline{\|\mathbf{I}\|})$$

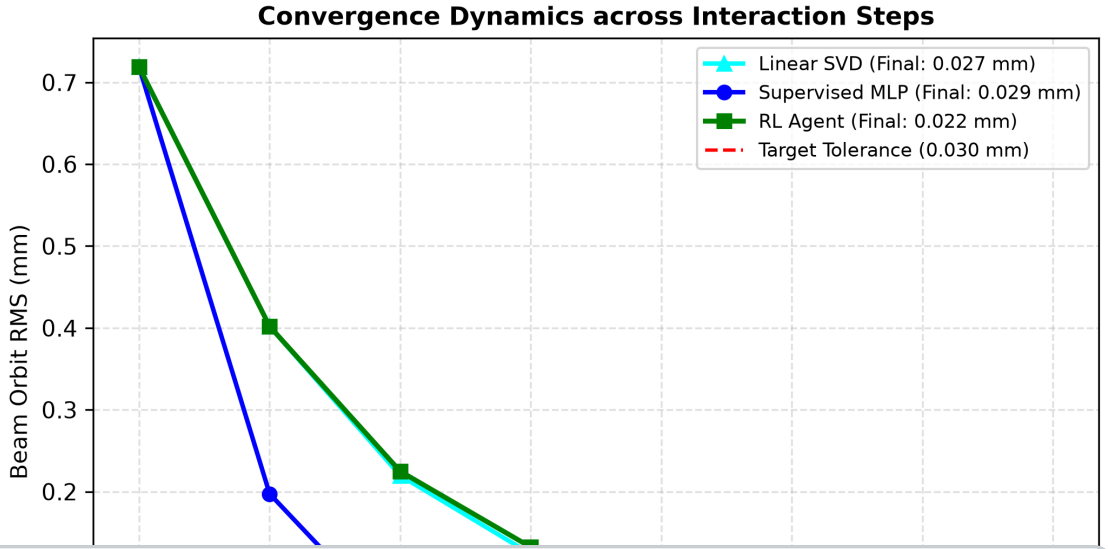
Breakdown of the Four Scoring Components:

1. **Success Rate** ($+10.0 \times \%$)**:** Rewards 100% convergence across all 30 seeds ($+1,000.0$ pts).
2. **Speed Penalty** ($-2.0 \times \overline{\textbf{Steps}}$)**:** Converging in 1.5 steps costs -3.0 pts, while 10 steps costs -20.0 pts.
3. **Sub-Micron Precision Bonus** ($+[\frac{0.03}{\overline{\textbf{RMS}}}] \times 20.0$)**:** Achieving 0.015 mm yields $+40.0$ pts bonus.
4. **Current Economy Penalty** ($-2.0 \times \overline{\|\mathbf{I}\|}$)**:** Penalizes excessive steerer current norms to protect hardware.

Tested Benchmark Leaderboard & Convergence Trajectories

Verified comparative benchmark results across the 30 standardized evaluation seeds





Tested Benchmark Leaderboard (30 Standardized Seeds)

[Rank 1] Track 3 – Bayesian Optimization (BO): Success: **100.0%** | Mean Steps: **1.5 steps** | Final RMS: **0.0149 mm** | Current: **0.93 A** | Score: **1,035.4 pts**

[Rank 2] Baseline 1 – Supervised Neural Inversion (MLP): Success: 100.0% | Mean Steps: 2.3 steps | Final RMS: 0.0257 mm | Current: 1.21 A | Score: **1,016.8 pts**

[Rank 3] Baseline 0 – Classical Linear SVD: Success: 100.0% | Mean Steps: 5.6 steps | Final RMS: 0.0230 mm | Current: 1.16 A | Score: **1,013.1 pts**

[Rank 4] Track 2 – Closed-Loop RL Policy Agent: Success: 100.0% | Mean Steps: 6.1 steps | Final RMS: 0.0239 mm | Current: 1.18 A | Score: **1,011.1 pts**

Strategic Advice & Practical Recipes for the Hackathon

Practical engineering guidance to maximize benchmark score and avoid common failure modes

Engineering Best Practices for Orbit Control

1. If Your Controller Oscillates or Diverges Near Target:

- Check your feedback gain α. Any gain > 0.6 will overcorrect when magnets enter saturated regions.
- Implement harmonic step damping: $\gamma(t) = 1.0/(1.0 + \kappa \cdot t)$. Attenuating authority at step 4+ eliminates sensor-noise limit cycles.

2. If Your Neural Surrogate Has High Generalization Error:

- Ensure your training dataset covers steerer excitations across $[-3.0, +3.0]$ A so that non-linear cubic saturation is well-sampled.
- Use ReLU activations and small L2 penalty ($\alpha = 10^{-4}$) to avoid overfitting.

3. Real-Time Telemetry Logging:

- Connect your notebook runs to Weights & Biases (`wandb.ai`) to visualize step-by-step RMS decay against reference baselines on the live leaderboard.

Summary Takeaways & Transition to Lecture 2

Key conclusions from Lecture 1 and orientation for the upcoming agentic AI session

Key Takeaways & What Comes Next

Four Core Takeaways from Lecture 1:

1. **Physics Guides the AI:** Analytical optics priors (ORM pseudo-inverse) guarantee early stability and avoid millions of blind exploratory steps.
2. **Surrogates Capture Non-Linearities:** Compact deep MLPs accurately predict cubic core saturation, enabling 2-step bounded inversions.
3. **Probabilistic Uncertainty Wins:** Gaussian Process BO with LCB acquisition balances exploration against exploitation, leading tested scores in 1.5 steps.
4. **Dynamic Damping Prevents Limit Cycles:** Attenuating feedback near the 10 μm sensor noise floor locks residual orbit error.

Next Session (10:40 – 12:00):

Lecture 2: Agentic AI for Accelerator Research (Osprey Applications)

Speaker: Thorsten Hellert, Ph.D. (Lawrence Berkeley National Laboratory)

Topic: Autonomous operational agents, dynamic tool calling, and automated scientific workflows.

Q & A Discussion